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EverFlavor AI

Capstone Project Proposal

A proposed conversational AI system that creates authentic global recipes, tracks nutrition, handles dietary restrictions, and finds specialty ingredients — all in one natural-language conversation.

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One Conversation. A Complete Meal Plan.

Users describe their situation in plain language — calories remaining, dietary restrictions, mood, and cuisine preference — and EverFlavor AI responds with a complete, safe, and actionable meal solution.

🗒 "I have 550 calories left, no pork, I'm tired after work, and I want something bold from Middle Eastern or Latin American cuisine. I can drive 15 minutes."

1

User Speaks

Natural language describing calories, restrictions, mood, cuisine preference, and travel radius

2

Agents Coordinate

Three specialized agents collaborate in real time to build a complete answer

3

System Delivers

- Complete step-by-step recipe from five major cuisine families
- Smart substitutions for missing ingredients
- Accurate calorie and nutrition information
- Nearest specialty supermarket within driving radius
- Full respect for all dietary restrictions

SYSTEM ARCHITECTURE

Three Specialized Agents, One Unified Response

EverFlavor AI uses a multi-agent architecture where each agent has a distinct role, dedicated memory, and access to purpose-built tools. Together, they cover the full journey from user intent to actionable meal plan.

Nutritionist & Preference Agent

Extracts remaining calories, all declared restrictions (religious, medical, ethical), preferred cuisine families, and driving radius. Builds a structured constraint profile passed downstream.

Chef Agent

Generates authentic recipes from five allowed cuisine families, applies calorie limits, produces practical substitutions when ingredients are missing. Output: complete recipe + substitution list.

Sourcing Agent

Queries Google Places API (geofenced to Houston metro area) and returns the nearest specialty supermarket likely to carry regional ingredients.

Four Groups. One Shared Struggle.

People want exciting, flavorful meals from different cultures — but everyday life gets in the way. Decision fatigue, missing ingredients, strict dietary rules, and zero time to research create a barrier that EverFlavor AI is designed to remove.



The Busy Professional

Works long hours under high stress. Sees eating mainly as a way to get energy. Plain chicken and rice feels boring, yet they have almost no energy left to cook after work.



The Impatient Foodie

Loves bold, exciting, and gourmet-style flavors. Has a packed schedule and very little free time. Dislikes bland frozen meals or basic home cooking, but lacks the hours needed to prepare and clean up complex recipes.



The Overworked Parent

Balances a career, household responsibilities, and childcare. Is tired of repeating the same simple kid-friendly meals. Wants more creative and interesting food for the family but usually settles for whatever is fastest.



The Efficiency-Focused Single or Student

Lives alone or in a shared/dorm space. Feels that cooking for one takes too much time and effort. Quickly gets bored of microwave ramen or fast food, but lacks kitchen confidence or free time to change the routine.

Grounded in Reliable, Public Data

EverFlavor AI does not rely on hallucinated facts or private datasets. Every nutritional claim, store location, and ingredient suggestion is anchored to verified, publicly available sources — ensuring accuracy and reproducibility.

Data Sources

- **Open Food Facts API** — Global nutritional database for calorie and nutrient lookup, ingredient lists, and packaged food details
- **USDA FoodData Central** — Authoritative U.S. government nutrition reference with standardized food and micronutrient data
- **Google Places API** — Real-world specialty supermarket locations by cuisine region, including addresses, hours, and nearby options
- **CrewAI Framework** — Multi-agent orchestration with tool-calling and memory for coordinating retrieval, reasoning, and response generation
- **Gradio** — Interactive chat interface for live class demonstrations and rapid prototype feedback

User Profile (Session-Based)

- Remaining daily and weekly calories
- Declared food restrictions: religious, medical, and ethical
- Preferred cuisine families (out of five: Asian, European, Latin American, African, Middle Eastern)
- Maximum driving distance for ingredient sourcing

Scope Boundaries

- Five major cuisine families only
- One metro area for the prototype (Houston)
- No custom model training required
- No large private datasets needed

Architecture Built for Accuracy and Safety

01

Multi-Agent Orchestration via CrewAI

Each agent has a clear role, dedicated memory, and access to specific tools. Sequential workflow: Nutritionist Agent extracts constraints → Chef Agent generates recipe → Sourcing Agent finds stores. Shared context prevents overlap and keeps the pipeline accountable from start to finish.

03

Smart Substitution Logic in the Chef Agent

When ingredients are missing, the Chef Agent suggests practical alternatives that preserve the dish's authenticity while still satisfying the user's nutritional goals and dietary constraints.

02

Nutrition Grounding Through Public APIs

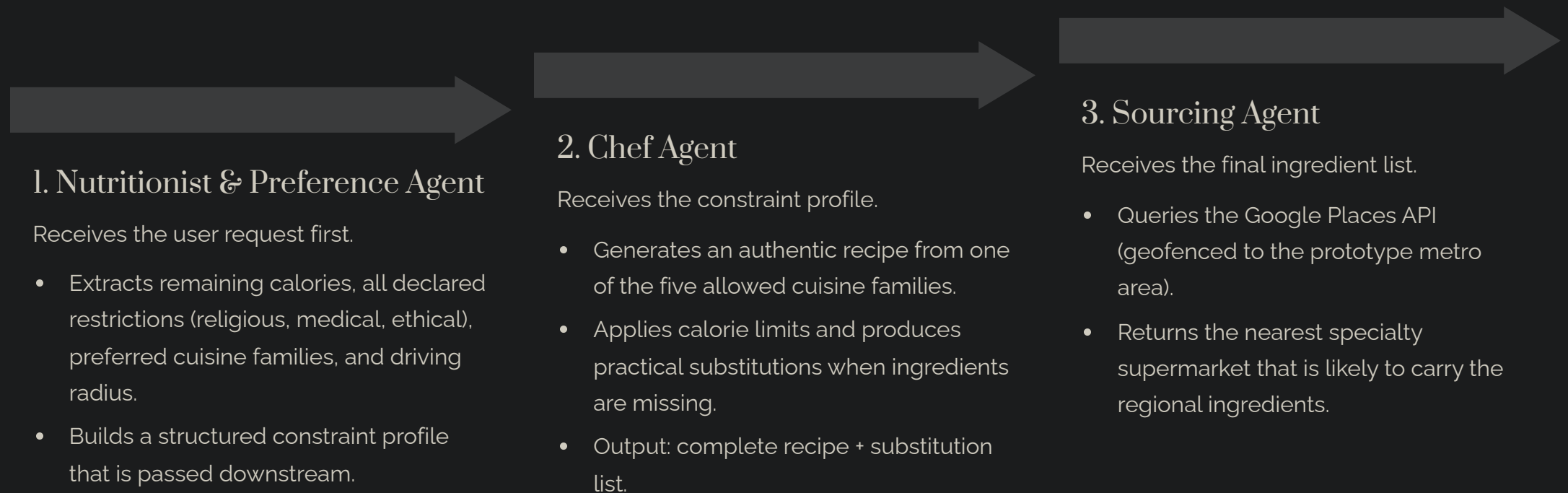
Calorie and nutrient data comes from Open Food Facts and USDA FoodData Central — not from the LLM's internal knowledge — keeping numbers verifiable, accurate, and reproducible across recipes.

04

Independent Safety Filter Layer

A hard-coded post-generation filter scans every ingredient and cannot be overridden by the language model — the final gate before any recipe reaches the user.

Orchestration Strategy: Sequential Workflow with Shared Context



Key Orchestration Features:

- Shared memory / context so every agent sees the same restrictions and calorie budget.
- Explicit task dependencies (Chef cannot start until Nutritionist finishes; Sourcing waits for Chef).
- Tool assignment is role-specific: Nutritionist → nutrition APIs + restriction checker, Chef → recipe generation + substitution logic, Sourcing → Google Places API.
- Independent safety filter runs after the Chef Agent finishes and before the final answer is shown to the user.

Human-in-the-Loop: Safety Gates & User Control

1

Safety Gate (Mandatory in Prototype)

After the Chef Agent generates a recipe and the automated filter runs, the system can pause and surface the recipe + restriction checklist for a human reviewer (team member or tester) before it is shown to the end user

This is especially important for allergy and religious-restriction cases

2

Substitution Review

When the Chef Agent proposes substitutions for missing ingredients, the user (or a reviewer) can accept, reject, or request alternatives in the chat interface

3

Preference Feedback Loop

After a meal suggestion is delivered, the user can give simple feedback ("too spicy", "perfect", "not authentic enough")

In later phases this feedback can be stored to improve future recommendations

4

Override Capability

The user always retains final control

They can ask the system to regenerate the recipe with stricter or looser constraints at any time

Note: In the Week 3-6 prototype we will implement the safety gate and basic user feedback. Full automatic learning from feedback is planned for later milestones if time allows. CrewAI's built-in human-feedback / human-in-the-loop mechanisms (Flow-based or webhook-based) will be used so the pause-and-review step is native to the orchestration.



Ingredient Interaction Knowledge: Three Tiers of Safety

Knowing what to combine and what never to combine is a distinct capability from generating a recipe. There are three tiers of safety and compatibility:

Tier 1: Hard Constraints (Never Learned)

IN SCOPE FOR PROTOTYPE

Some combinations are unsafe rather than merely unpleasant. No learned component may ever decide they are acceptable. These live in the same non-overridable filter layer as the allergen checks.

- Cross-contamination between raw poultry/ground meat and ready-to-eat items
- Prolonged contact between acidic/salty foods and reactive cookware
- Minimum internal temperature floors for poultry, ground meat, and eggs

Implemented as hard rules, not as model behavior.

Tier 2: Chemistry and Physics (Encoded as Rules)

IN SCOPE FOR PROTOTYPE

These combinations are not dangerous; they simply fail in predictable ways. A rule base handles them more reliably than a learned model. The Chef Agent consults this rule set before finalizing a recipe.

- Raw pineapple/papaya/kiwi prevent gelatin from setting
- Acid added directly to dairy curdles it
- Salt draws water from vegetables by osmosis
- Baking soda requires an acid
- Cornstarch breaks down under sustained heat
- Maillard browning requires a dry surface above 140°C

Important: Almost every rule is conditional rather than absolute (e.g., raw pineapple blocks gelatin, but canned pineapple does not).

Tier 3: Flavor Compatibility (Learnable)

PLANNED EXTENSION

This is where a learned model earns its place. Uses published food pairing research and co-occurrence mining from recipe corpus.

- Would improve substitution logic by learning substitutability from context
- Bias risk: Co-occurrence learns what is common, not what is good. Must stratify by cuisine and evaluate per cuisine.

Design principle: Tiers 1 and 2 may reject a recipe. Tier 3 may only reorder suggestions.

Long-Term Vision: From Recipes to Robotic Cooking

This is a future extension well beyond the 16-week course, but shows where the architecture leads.

1

Key Insight: Manipulation vs. Timing

- The common assumption is that the robotic arm is the central challenge
- It is difficult, but it is not where the meaningful AI work sits
- Food is deformable, variable, and slippery
- Timing is the real modeling problem
- Cooking is a scheduling problem under resource constraints with irreversible actions
- Limited burners, limited pans, limited hands
- The binding constraint is that everything should finish hot at the same moment
- This is a temporal planning problem, and one where a system can be reliably better than a home cook

2

Stove Understanding as System Identification

- Gas and electric are not cosmetic variations; they are different control problems
- Gas responds almost immediately
- Electric coil carries substantial thermal lag (30–90 seconds of stored heat)
- Induction responds quickly but heats only ferrous cookware
- The approach is to learn a thermal response model for each hob and pan combination using standard system identification techniques

3

Prototypable Slices (in order of increasing hardware risk)

1. Coaching mode (no actuation): A camera and temperature probe observe a human cook and advise when to flip, when the pan is too cold, and when to add the next ingredient
2. Instrumented single hob: Thermocouple or infrared sensor + camera + controlled induction hob that holds a target temperature through disturbances
3. One degree of manipulation freedom: A single rotating stirring arm in a pot (useful for risotto, roux, and anything prone to scorching)

4

The Bridge from the Current System

- Our present system outputs a recipe (an unordered list of instructions)
- A robotic system requires a temporal plan — a schedule of parallel tasks with dependencies and durations
- Converting recipe text into a structured task graph is the bridge between the two
- That conversion is useful on its own, even with no robot involved, because it would let the current system tell a user what to start first and what to do while waiting
- It is the one piece of this extension that could be prototyped within the course timeline if the schedule allows

How We Measure What Works

Each metric is designed to be measurable, testable, and meaningful — reflecting real user needs rather than abstract benchmarks. These targets define what "done and working" looks like for the prototype.

5%

Nutritional Accuracy

Calorie calculations must stay within a 5% margin of error compared to manual calculations using the same data sources.

100%

Restriction Compliance

Every recipe shown to the user must pass the automated filter for all declared religious, medical, and ethical restrictions.

0

Hallucinated Stores

The Sourcing Agent must return only real, verifiable specialty stores. No fabricated names or addresses are acceptable.

1

Conversational Turn

For realistic queries from busy professionals, impatient foodies, overworked parents, and efficiency-focused singles/students, the system must deliver a complete, usable answer in a single conversational exchange.

Known Risks and How We Address Them

Risk	Impact	Likelihood	Mitigation Strategy
LLM forgets a food restriction	High	Medium	Hard-coded post-generation filter that cannot be overridden by the language model
Hallucinated store locations	Medium	Medium	Strict use of Google Places API with result validation — no LLM-generated addresses
Inaccurate calorie counts	Medium	Low-Medium	All calculations grounded in Open Food Facts and USDA data, not LLM estimates
Scope creep	High	Medium	Strict limits: 5 cuisine families, one metro area, defined substitution logic
API rate limits or unexpected costs	Low-Medium	Low	Use free tiers, implement caching, and limit demo query volume
Poor quality substitutions	Medium	Medium	Test and refine substitution logic with clear criteria for dish recognition and satisfaction

Safety Is Architecture, Not a Promise

Recommending food that violates a user's religious beliefs, triggers an allergy, or conflicts with a medical condition can cause real harm. EverFlavor AI treats safety as a structural requirement — not something left to the language model's memory.

Structured Restriction Profiles

All restrictions are stored in a structured user profile — not just in chat history — ensuring they persist and are checked on every single recipe generated.

Independent Post-Generation Filter

After the Chef Agent creates a recipe, an automated filter scans every ingredient. Any conflict causes the recipe to be rejected or rewritten before the user sees it.

Full Restriction Coverage

Supports religious restrictions (Kosher, Halal, Hindu, Buddhist, Jain, Catholic), health & medical restrictions (allergies, Celiac, lactose intolerance, diabetes), and ethical choices (vegetarian, vegan).

Transparent Disclaimers

Clear messaging that EverFlavor AI is a planning tool — not medical or religious advice. Calorie calculations are transparent so users can verify the numbers themselves.

SUMMARY

A Real System for Real People

EverFlavor AI brings together multi-agent intelligence, global cuisine knowledge, practical substitutions, specialty-store sourcing, and a rigorous safety architecture — all designed to fit within a 16-week course timeline and deliver a working, demonstrable prototype.

Global Reach

Five major cuisine families: Asian, European, Latin American, African, and Middle Eastern — with authentic recipes and real specialty stores.

Safety First

A hard-coded, non-overridable filter ensures no recipe ever violates a user's declared religious, medical, or ethical restrictions.

One Conversation

From question to complete meal plan — recipe, substitutions, nutrition, and sourcing — in a single conversational turn.

Focused Scope

Deliberately limited geography and cuisine range ensures the team delivers a reliable, polished system within the course timeline.

 **Team EverGlow** — Chloe Tu · Eva Abou Abou Harb · Francisco Medinadiaz | ITAI 2277 · Houston City College · Professor Anna Devarakonda

16-Week Project Timeline: EverFlavor AI

Our 16-week timeline is structured to deliver a functional and thoroughly tested prototype within the academic semester, balancing rapid development with rigorous quality and safety checks.

Weeks 1-3: Research & Architecture

Team formation, problem finalization, and project proposal. Tools setup (GitHub, shared Drive). Final problem statement and user personas locked.

Weeks 9-13: Testing, Safety Filter, Refinement

Full sequential orchestration of all 3 agents. Implement hard-coded safety filter. Human-in-the-loop with safety gate. Comprehensive testing against success metrics.

1

2

3

4

Weeks 4-8: Agent Development & API Integration

Build Nutritionist, Chef, and Sourcing Agents. Integrate Open Food Facts & USDA APIs. Develop and test substitution logic. Connect Google Places API (limited to Houston metro).

Weeks 14-16: Demo Prep & Final Polish

Documentation and architecture diagrams. Full end-to-end testing. Peer feedback collection. Live Gradio demo and final presentation.

EverFlavor AI in Action: From Query to Cuisine

Watch EverFlavor AI seamlessly integrate multi-agent intelligence to deliver a personalized meal plan based on a user's unique needs, restrictions, and preferences. Here's a typical interaction:

User Query:

"I have 500 calories left for dinner. I'm allergic to peanuts and need a kosher meal. What can I make that's Asian-inspired?"

The user provides caloric goals, dietary restrictions, and cuisine preferences in a natural language prompt.

EverFlavor AI's Multi-Part Response:

1

🍳 RECIPE

Chef Agent:
"Quick Chicken & Broccoli Stir-Fry (Peanut-Free)"

- Marinated chicken breast, broccoli florets, carrots, bell peppers.
- Sauce: soy sauce (kosher), ginger, garlic, sesame oil, cornstarch.
- Served over a small portion of jasmine rice.

2

🥗 NUTRITION

Nutrition Agent:

- Calories: 485 kcal
- Protein: 42g, Carbs: 38g, Fat: 20g
- [Source:](#) USDA & Open Food Facts data, not LLM estimates.

3

🏪 SOURCING

Sourcing Agent:

- Find kosher chicken at **Kosher Mart** (123 Main St, Houston).
- All other ingredients available at **SuperFresh Market** (456 Oak Ave, Houston).